

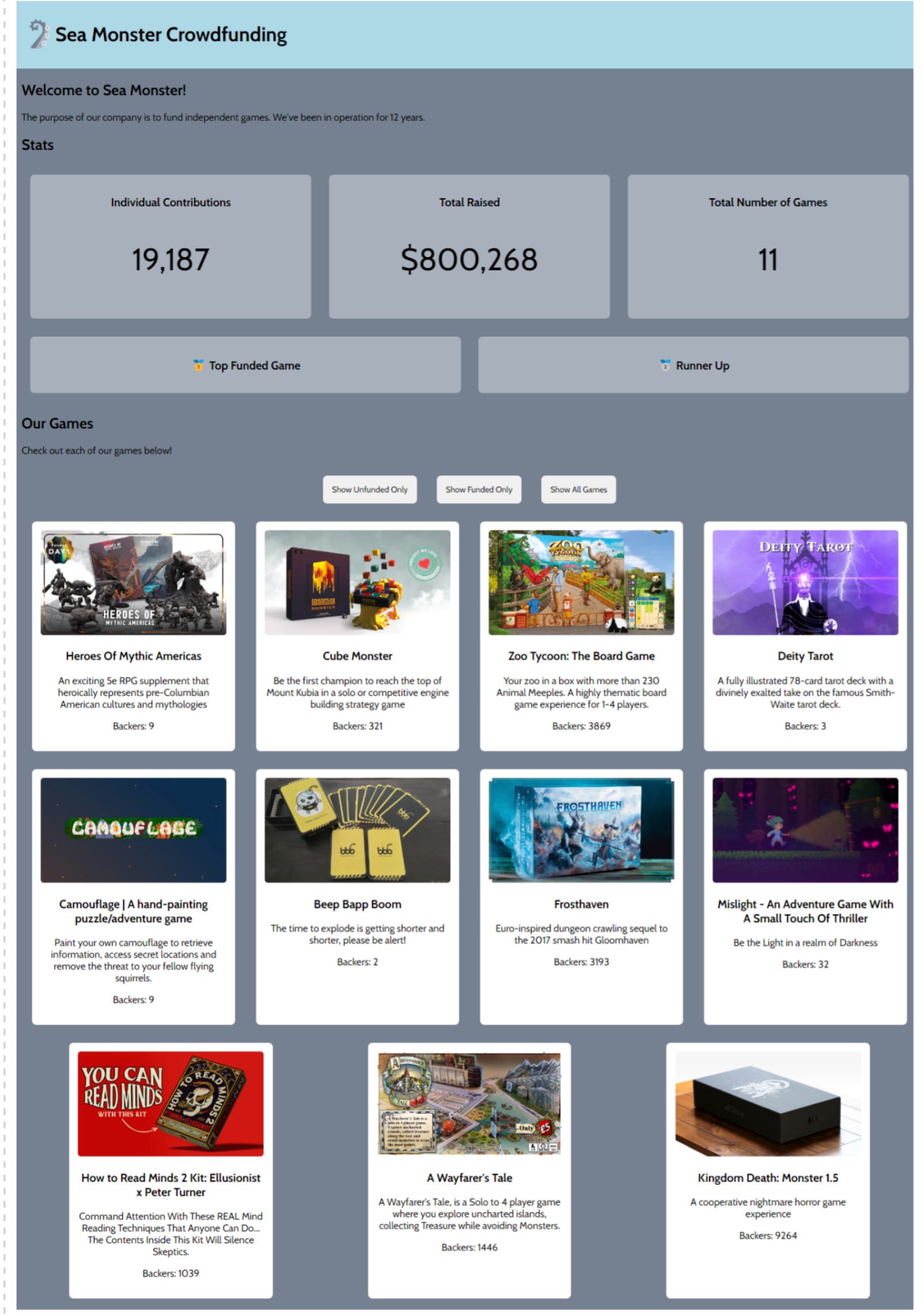
WEB102 Prework

Challenge 5 of 7: Add functions to filter the funded and unfunded games

✔ Checkpoint

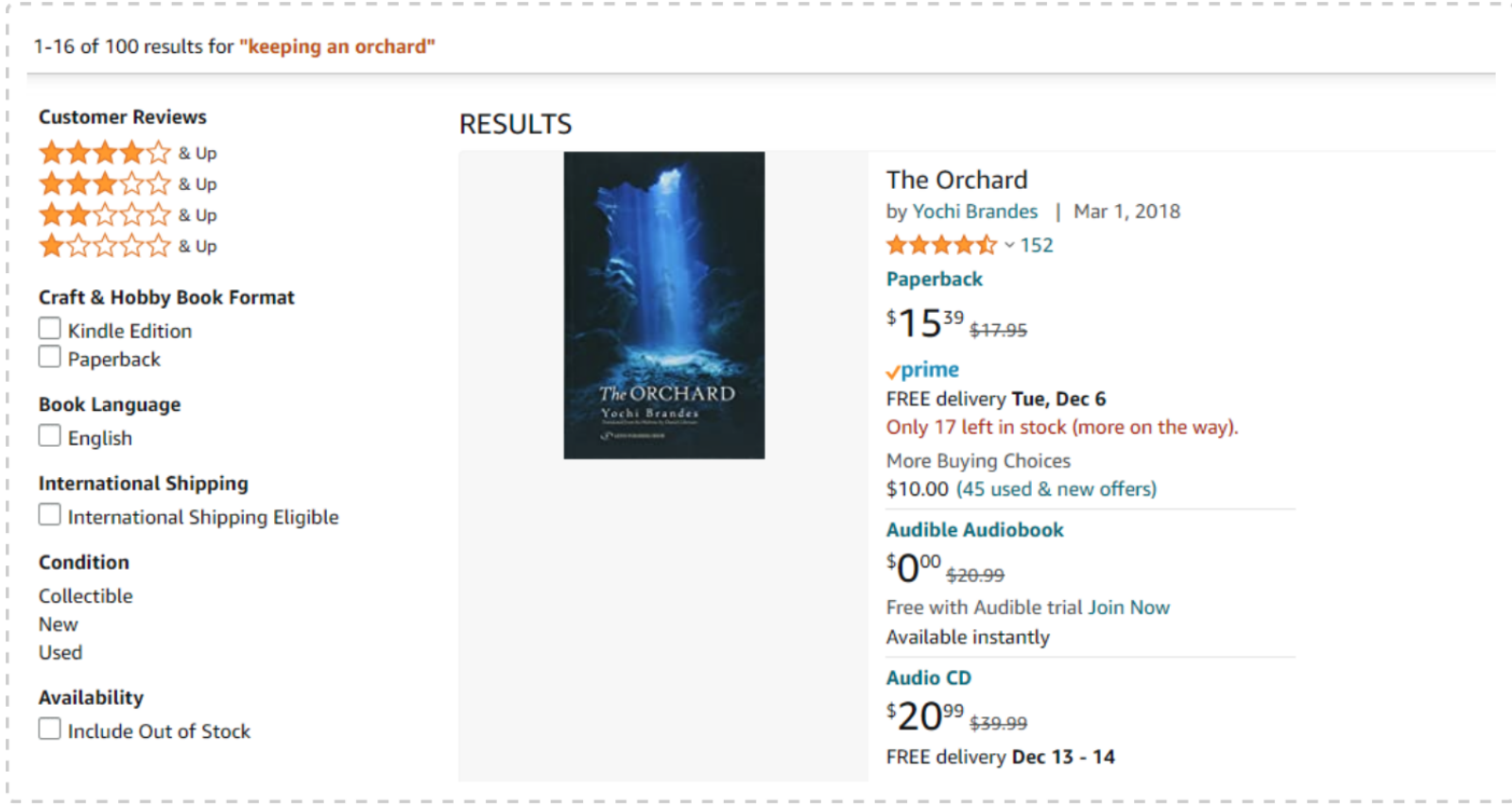
Congrats and welcome to Challenge 5! You're well on your way to completing the prework.

At this point, your website should look something like the screenshot below. Compare your work with the exemplar and see if there are any changes you might need to make!



🖌 Background

You've probably encountered a ton of filters in the wild. When searching for a book on Amazon, for example, the website displays a list of filters on the lefthand side to help you tailor the results by rating, condition, availability, and more:



So

how can we accomplish filtering quickly and easily?

JavaScript's built in function **filter** comes in handy here. Unlike **reduce**, which is designed to collapse a list into a single value, **filter** is designed to return a list of elements that match a given condition. Filter takes only one argument: a callback that returns **true** when an item should be added to the new list and **false** when it should be excluded.

Imagine we're back in our office at Spotify, working on another feature for Spotify Wrapped. This time, we want to display a list of songs that the user played more this year compared with last year. How can we take a list of **songs** and construct a new list of only those songs with higher values this year?

Take a look at this JavaScript code that accomplishes this task:

```
let listOfIncreasedSongs = songs.filter ( (song) => {  
  return song.playsIn2022 > song.playsIn2021;  
});
```

If you want to read up some more on the **filter** function, check out these resources:

- [Array.prototype.filter\(\)](#)
- [JavaScript Array filter: Filtering Elements](#)

Let's use our new tool **filter** to complete some challenges!

🎯 Challenge Steps

Navigate to the Challenge 5 section of **index.js** to begin.

➡➡➡ Step 1

First, complete the **filterUnfundedOnly** function. Use JavaScript's **filter** function to only show games whose amount raised is less than the goal amount.

- Use the **addGamesToPage** function you previously wrote to add the games to the page

🔑 **Secret Key component 1:** How many games are in the array returned by **filterUnfundedOnly**? (Hint: use **console.log** or call the function to see the result.)

➡➡➡ Step 2

Next, complete the **filterFundedOnly** function to only show games whose amount raised is greater than or equal to their goal amount.

- Use **addGamesToPage** to display the games

🔑 **Secret Key component 2:** How many games are in the array returned by **filterFundedOnly**? (Hint: use **console.log** or call the function to see the result.)

➡➡➡ Step 3

Next, finish the **showAllGames** function so that all games are displayed.

➡➡➡ Step 4

Finally, create an event listener for each button and connect each event handler with the correct button.

- For example, connect the **filterUnfundedOnly** function to the **unfundedBtn**.
- If you need a refresher on connecting event handlers to DOM elements, [check out this tutorial](#).

🔑 **Secret Key component 3:** What would happen if the **deleteChildElements** function was never called within the three event handlers you wrote?

1. The buttons would throw an error, causing the website to become unresponsive to user inputs. (keyword: CORDUROY)
2. The buttons would show only the full list of games instead of the filtered lists. (keyword: LINEN)
3. The buttons would delete all their elements at the end of the function instead of the beginning, failing to display the games. (keyword: WOOL)
4. The buttons would always add their list of games to the games already displayed, creating a growing list of games. (keyword: FLANNEL)

🔑 **Secret Key component 4:** Which type of event does each button listen for?

🚀 Moving on to Challenge 6

Create the 🔑 **Secret Key** from its components with no spaces in between:

- Component 1: How many games are in the array returned by **filterUnfundedOnly**?
- Component 2: How many games are in the array returned by **filterFundedOnly**?
- Component 3: What would happen if the **deleteChildElements** function was never called within the three event handlers you wrote?
- Component 4: Which type of event does each button listen for?

Hint: your 🔑 **Secret Key** should be 14 characters long.

Reminders about 🔑 **Secret Keys**:

- Keys for the Offline PDF version of prework **are case sensitive!**
 - (For example, "SEA_MONSTER" is NOT the same as "sea_monster")
- Secret keys can be any combination of letters, numbers, and symbols.

Use your 🔑 **Secret Key** to unlock the next PDF file. If your 🔑 **Secret Key** is correct, you will move onto the next step!